The Sleep-Activity Trade-Off (Pre-COVID Baseline)

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Introduction

Welcome to my Google Analytics Certificate Capstone project. Below you will find my data analysis process for Case Study 2: Analyzing smart device data as a junior data analyst for Bellabeat.

Business Context

Bellabeat, a high-tech manufacturer of health-focused products for women, faces a critical strategic question: How can they leverage insights from smart device usage data to inform their marketing strategy and product development?

Co-founder Urška Sršen tasked the marketing analytics team with analyzing real-world usage patterns to answer this question. This case study responds to that challenge by examining FitBit Fitness Tracker data from 33 participants, focusing on the relationship between daily activity and sleep behavior.

A Critical Temporal Note: This data was collected in April-May 2016—nearly four years before the COVID-19 pandemic fundamentally altered how we work, exercise, and sleep. The patterns revealed here represent pre-pandemic "normal," making them a valuable baseline for understanding how health behaviors have evolved in our post-COVID world.

Personal Motivation: Why This Analysis Matters

I relate very well to the women in this dataset. As a half marathon runner and cyclist, I understand the constant balancing act of fitting exercise into an already packed schedule of work, family, and personal commitments. I've been the person who sets a 5:30 AM alarm for a training run despite getting to bed past midnight. I've powered through workdays on 5 hours or less of sleep because "the miles had to get done."

That's why I would benefit from a fitness device that notified me when I was sacrificing sleep for activity a tool that encouraged balance, not just achievement. Even as someone who's health-conscious and active, I can do better to balance exercise, work, family, and my personal well-being.

This personal connection shapes my analytical lens. I'm not examining abstract data points; I'm investigating patterns that affect real women's daily lives, including my own.

The Bigger Picture: Fitness Technology's Evolution

The trends I uncover in this analysis—particularly around the sleep-activity trade-off represent the kinds of actionable insights that can transform how we think about fitness technology. This work is directly relevant to fitness tracker industry leaders like **Garmin**, **Fitbit**, and **Apple**, companies that are shaping the future of how athletes and fitness enthusiasts track and improve their performance.

The COVID-19 Factor: What makes this analysis particularly intriguing is its temporal positioning. This 2016 data captures behavior patterns before:

- Remote work became widespread (eliminating commutes, changing daily activity patterns)
- Gym closures drove outdoor running and home workouts to surge
- "Zoom fatigue" and screen time skyrocketed
- Mental health awareness around sleep hygiene intensified
- Work-life boundaries blurred with always-on remote culture

The running and fitness community needs technology built by people who understand both the data science *and* the lived experience of trying to stay healthy in the real world. That dual perspective—amplified by understanding how global events reshape health behaviors—drives everything in this analysis.

Methodology Overview

This case study employs a rigorous analytical approach:

Data Sources:

- FitBit Daily Activity Data (n=33 participants, 940 observations, April-May 2016)
- FitBit Sleep Data (n=24 participants, 413 observations)
- NHANES 2009-2011 (n=5,020 US women, for validation)

Temporal Context:

- Data collected pre-COVID (April-May 2016)
- Represents baseline "normal" activity and sleep patterns
- Provides foundation for future comparative analyses with post-2020 data

Analytical Steps:

1. Exploratory data analysis of activity, sleep, and sedentary behavior
2. Data cleaning and quality checks (including duplicate detection and removal)
3. Dataset integration (merging sleep and activity by participant and date)
4. Correlation analysis to quantify the sleep-activity relationship
5. Validation against nationally representative NHANES data
6. Translation of findings into strategic marketing recommendations

Central Research Question:
Is there a relationship between sleep duration and daily activity levels? Do active women sacrifice sleep for movement, or do the two support each other?

Future Research Direction:
How have these patterns changed in the post-COVID era, and what do those changes mean for fitness technology strategy?

The answer to the first question, as you'll see, has significant implications for how we design and market wellness technology. The answer to the second question awaits data—and represents exactly the kind of longitudinal analysis I'm eager to pursue professionally.

Let's dig in.

Data and Initial Exploration

Datasets

- **FitBit Daily Activity**: 33 participants, 940 observations
- **FitBit Sleep Data**: 24 participants, 413 observations

- **FitBit Weight Log**: 8 participants, 67 observations
- **NHANES Physical Activity** (comparison): 5,020 female participants

Understanding the Sample Size Differences

Why do participant counts differ across datasets?

The FitBit data comes from different tracking behaviors:

- **33 participants** tracked daily activity (steps, calories, sedentary time)
- **24 participants** tracked sleep (not everyone used the sleep tracking feature)
- **8 participants** logged weight (manual entry, lowest adoption)

This reflects real-world usage patterns: activity tracking is passive and automatic, while sleep tracking requires wearing the device to bed (not everyone does), and weight logging requires manual effort (lowest engagement).

Impact on Analysis:

- Individual dataset analyses use all available data for that metric
- Merged sleep + activity analysis uses only the **24 participants who tracked both**
- This is appropriate because we're examining the *relationship* between sleep and activity, which requires both data points

Dataset Summary

Dataset	Participants (n)	Observations	Usage Pattern
Daily Activity	33	940	High (automatic tracking)
Sleep Tracking	24	413	Medium (requires wearing to bed)
Weight Log	8	67	Low (manual entry required)
Merged Sleep + Activity	**24**	**410**	**Paired measurements only**

Key Insight: Feature adoption decreases as effort required increases. This informs product strategy—passive tracking gets highest engagement.

Summary Statistics

Daily Activity Dataframe

Statistic	TotalSteps	TotalDistance	SedentaryMinutes
Min.	0	0.000	0.0
1st Qu.	3790	2.620	729.8
Median	7406	5.245	1057.5
Mean	7638	5.490	991.2
3rd Qu.	10727	7.713	1229.5
Max.	36019	28.030	1440.0

Sleep Dataframe

Statistic	TotalSleepRecords	TotalHoursAsleep	TotalTimeInBed
Min.	1.000	0.9667	61.0
1st Qu.	1.000	6.0167	403.0
Median	1.000	7.2167	463.0
Mean	1.119	6.9911	458.6
3rd Qu.	1.000	8.1667	526.0
Max.	3.000	13.2667	961.0

Weight Dataframe

Statistic	WeightPounds	Fat	BMI
Min.	116.0	22.00	21.45
1st Qu.	135.4	22.75	23.96
Median	137.8	23.50	24.39
Mean	158.8	23.50	25.19
3rd Qu.	187.5	24.25	25.56
Max.	294.3	25.00	47.54
NA's		65	

NHANES Dataframe (Females Only)

Statistic	SleepHrsNight	BMI
Min.	2.000	12.88
1st Qu.	6.000	21.24
Median	7.000	25.56
Mean	6.991	26.77
3rd Qu.	8.000	31.20
Max.	12.000	81.25
NA's	1079	179

Initial Insights from FitBit Data

Activity Patterns

Key Finding: Despite taking over 7,000 steps daily (median), participants average 17.6 hours of sedentary time. This reveals that daily step counts and sedentary behavior are somewhat independent - users may exercise intentionally but then spend long periods sitting.

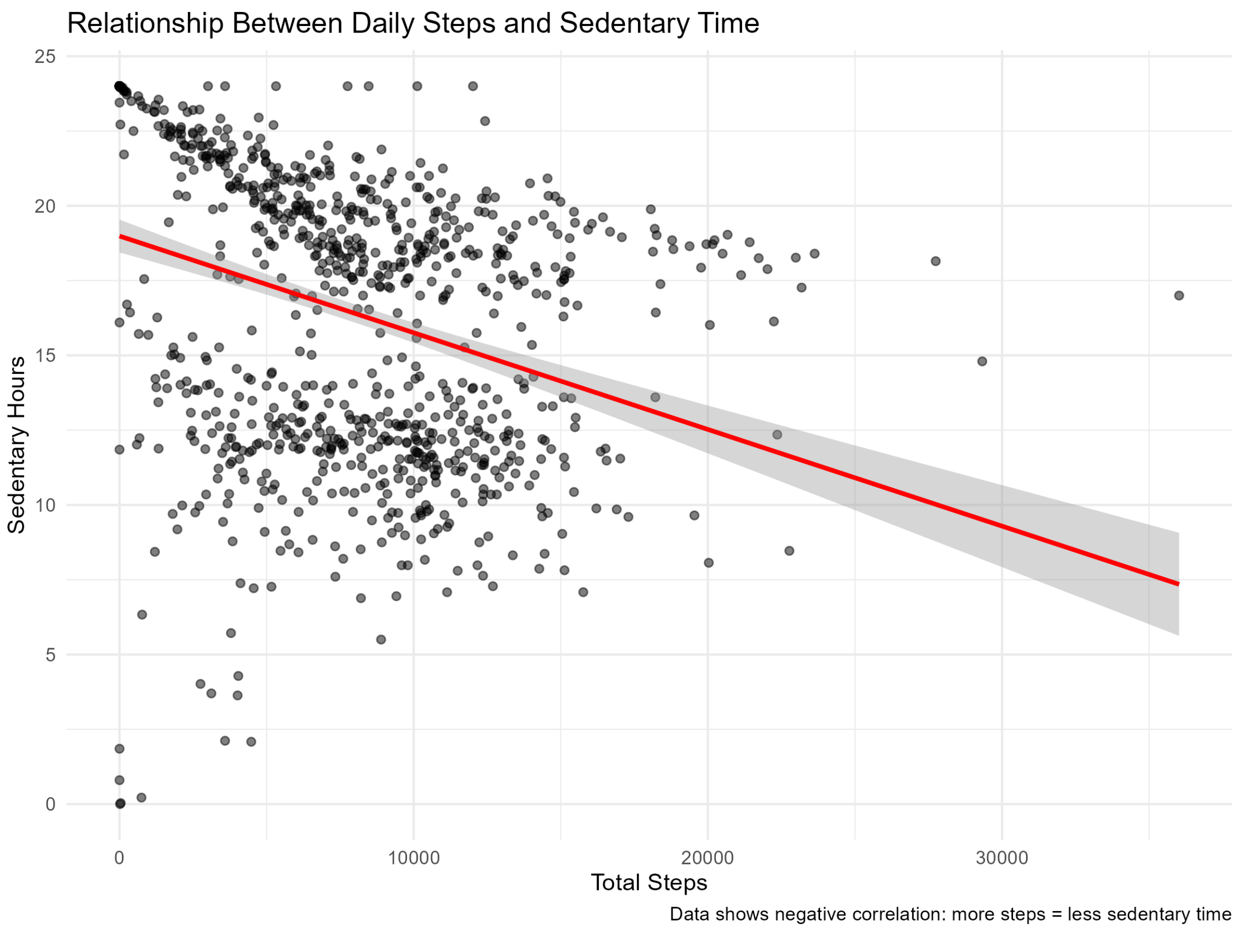
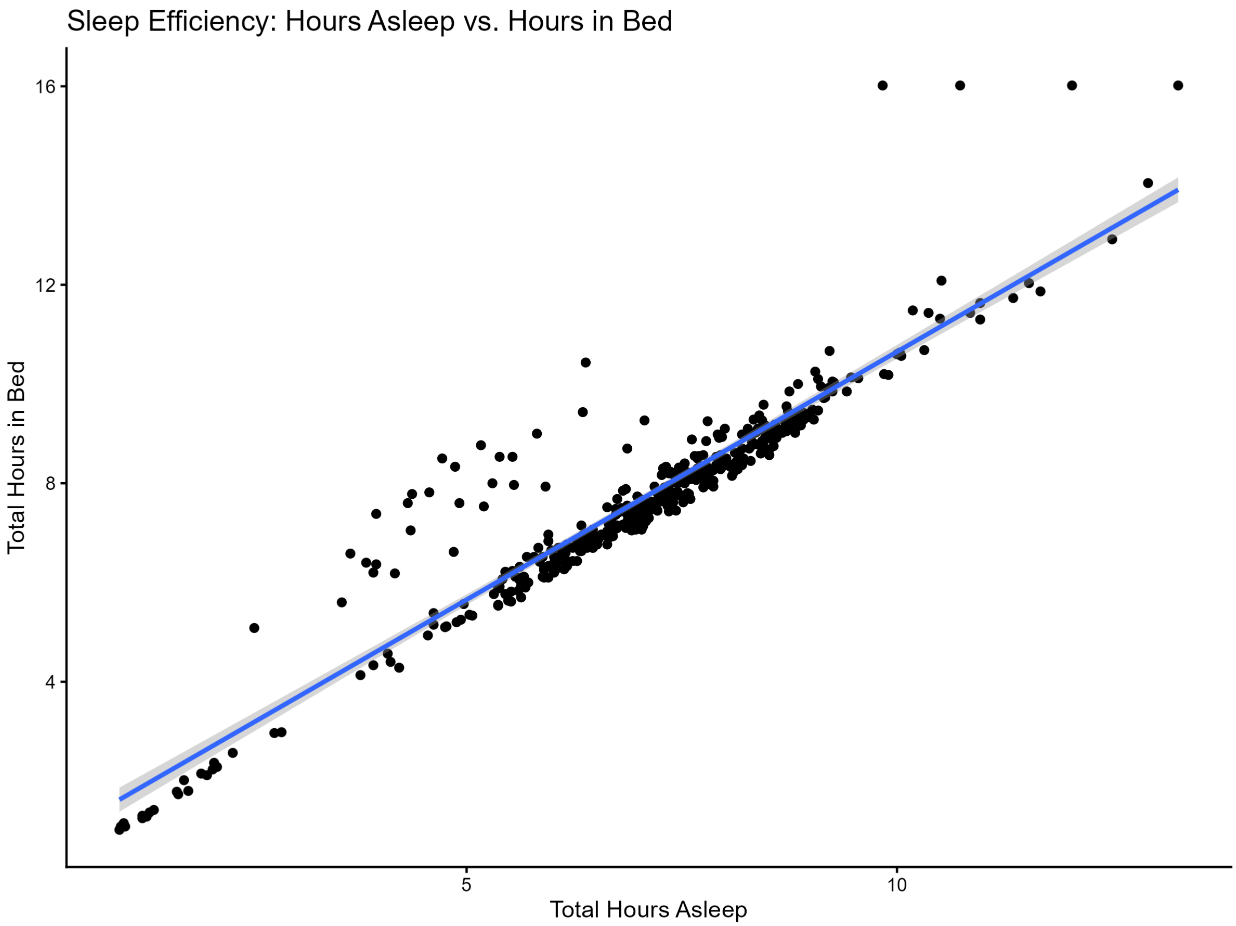
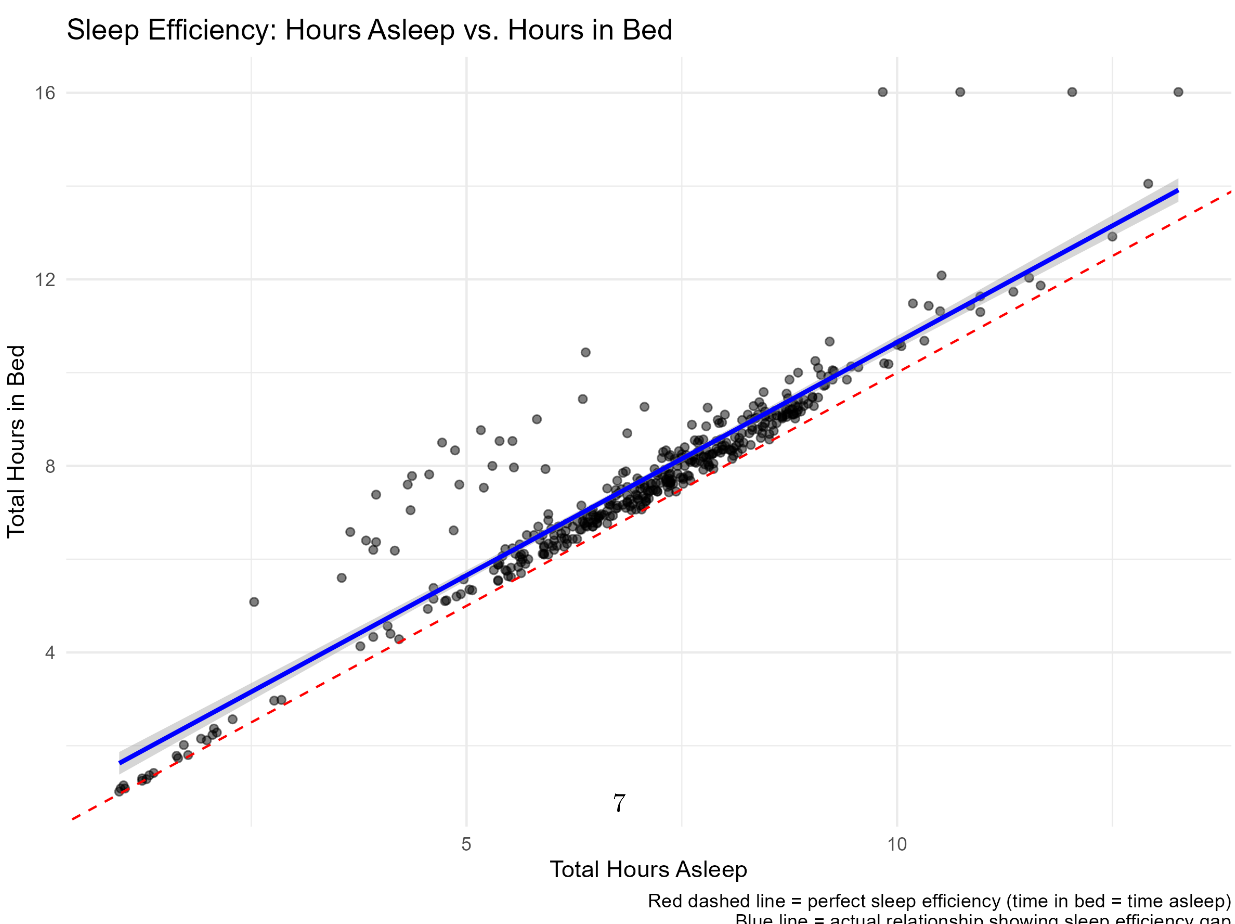


Figure 1: Scatter plot: Daily Steps vs Sedentary Time

The scatter plot above shows a negative correlation between daily steps and sedentary minutes. Users who take more steps throughout the day tend to spend fewer minutes sedentary. However, even highly active users (15,000+ steps) still accumulate 13-20 hours of sedentary time daily, suggesting that exercise alone may not eliminate prolonged sitting.

Sleep Efficiency





Key Finding: While there's a strong positive correlation between time in bed and actual sleep (R^2 = 0.866), the gap between the red line (perfect efficiency) and blue line (actual relationship) indicates time spent awake in bed. Users could benefit from sleep quality improvements, not just duration tracking.

The Sleep-Activity Trade-Off: A Critical Discovery

To understand if sleep and activity compete for time in users' lives, I merged the sleep and activity data sets by participant ID and date.

Data Quality Checks

Duplicate Detection:

Before proceeding with analysis, I checked for duplicate records that could skew results.

Method:

- Checked for duplicate entries by participant ID and date
- Identified 3 duplicate sleep records and 543 duplicate activity records

Action Taken:

- Removed all duplicates using `distinct()` function
- Verified removal with post-cleaning duplicate checks

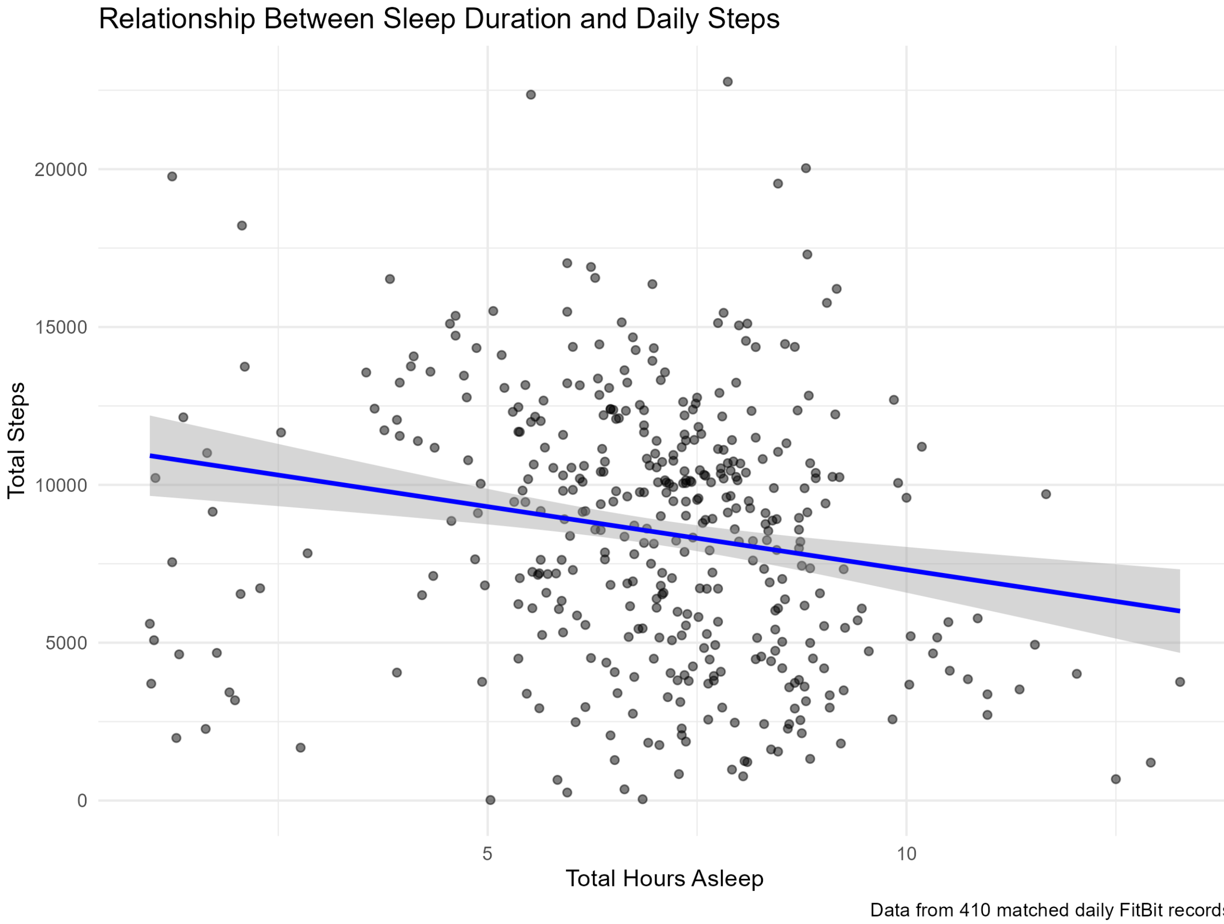
Results: 410 matched daily records from 24 participants with both sleep and activity data.

Dataset	Duplicates Found	Records After Cleaning
Sleep Data	3	410
Activity Data	543	940

Data quality insight:

Of the 413 unique sleep records, 410 (99.3%) successfully matched with daily activity data, indicating high data quality and consistent tracker usage among participants who engage with sleep tracking. However, only 410 of the 940 daily activity records (43.6%) included corresponding sleep data, suggesting that while users consistently track daily activity, sleep tracking adoption is lower. This presents an opportunity for Bellabeat to encourage more comprehensive sleep monitoring.

Correlation Analysis



From the scatter plot:

- Clear downward trend line
- People sleeping 8+ hours average around 6,000-8,000 steps
- People sleeping <6 hours average around 9,000-12,000 steps
- Wide variation exists, but the pattern is consistent

Statistical Results:

- Correlation: **-0.19** (negative correlation)
- P-value: **0.0001054** (highly statistically significant)
- **Interpretation**: People who sleep MORE tend to take FEWER steps, and vice versa

Segmented Analysis:

- Poor Sleep (<6 hrs): Median ~10,000 steps (HIGHEST activity!)
- Moderate Sleep (6-8 hrs): Median ~9,500 steps
- Good Sleep (8+ hrs): Median ~6,500 steps (LOWEST activity)

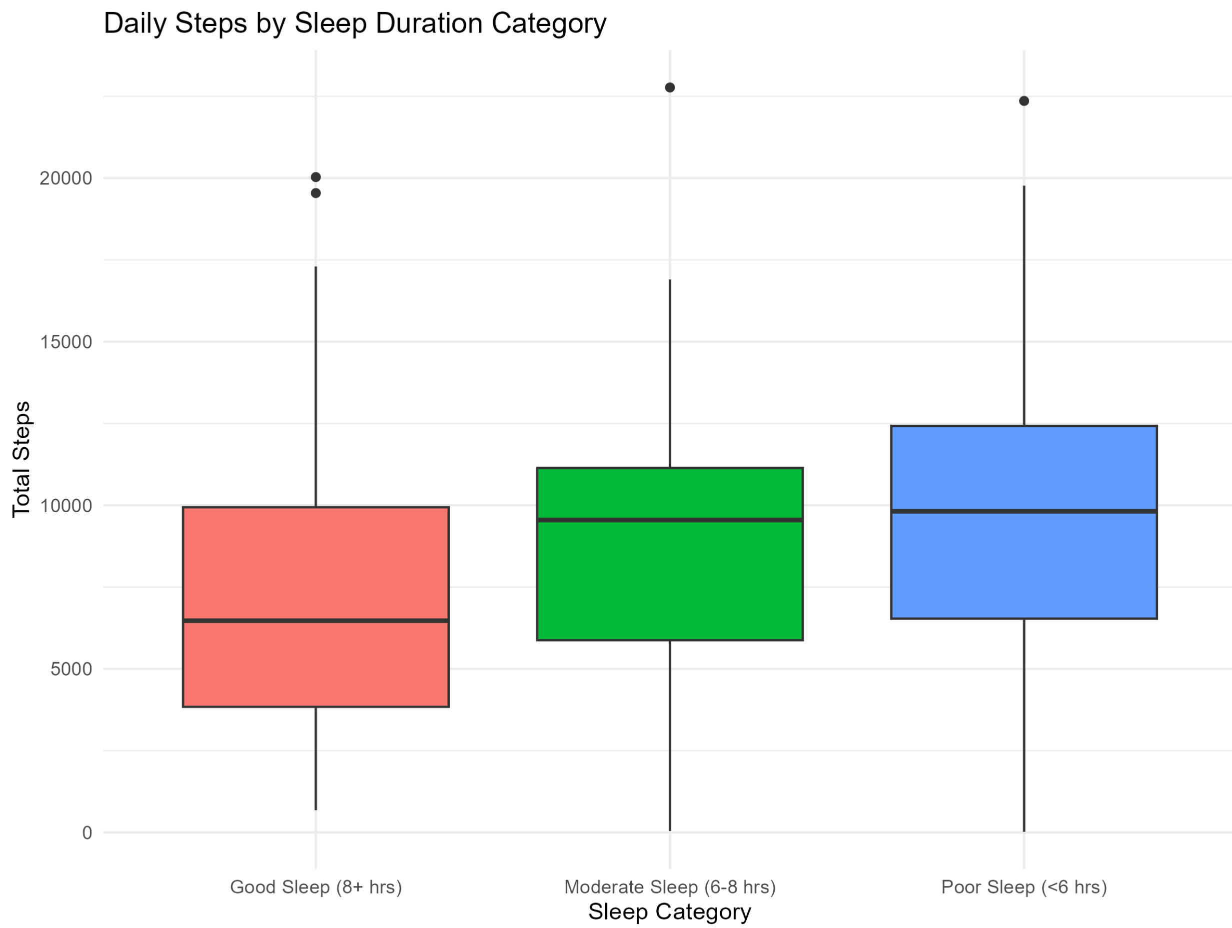


Figure 2: Boxplot: Daily Steps by Sleep Category

What This Means

This negative correlation suggests busy, active women are **sacrificing sleep for activity**. The most active users are also the most sleep-deprived. This is unsustainable and presents a key opportunity for Bellabeat to position itself as a balance coach, not just an activity tracker.

Validating Against NHANES: Is Our Sample Representative?

Despite the small FitBit sample (n=24), validating against nationally representative data strengthens our conclusions.

Sleep Duration Comparison

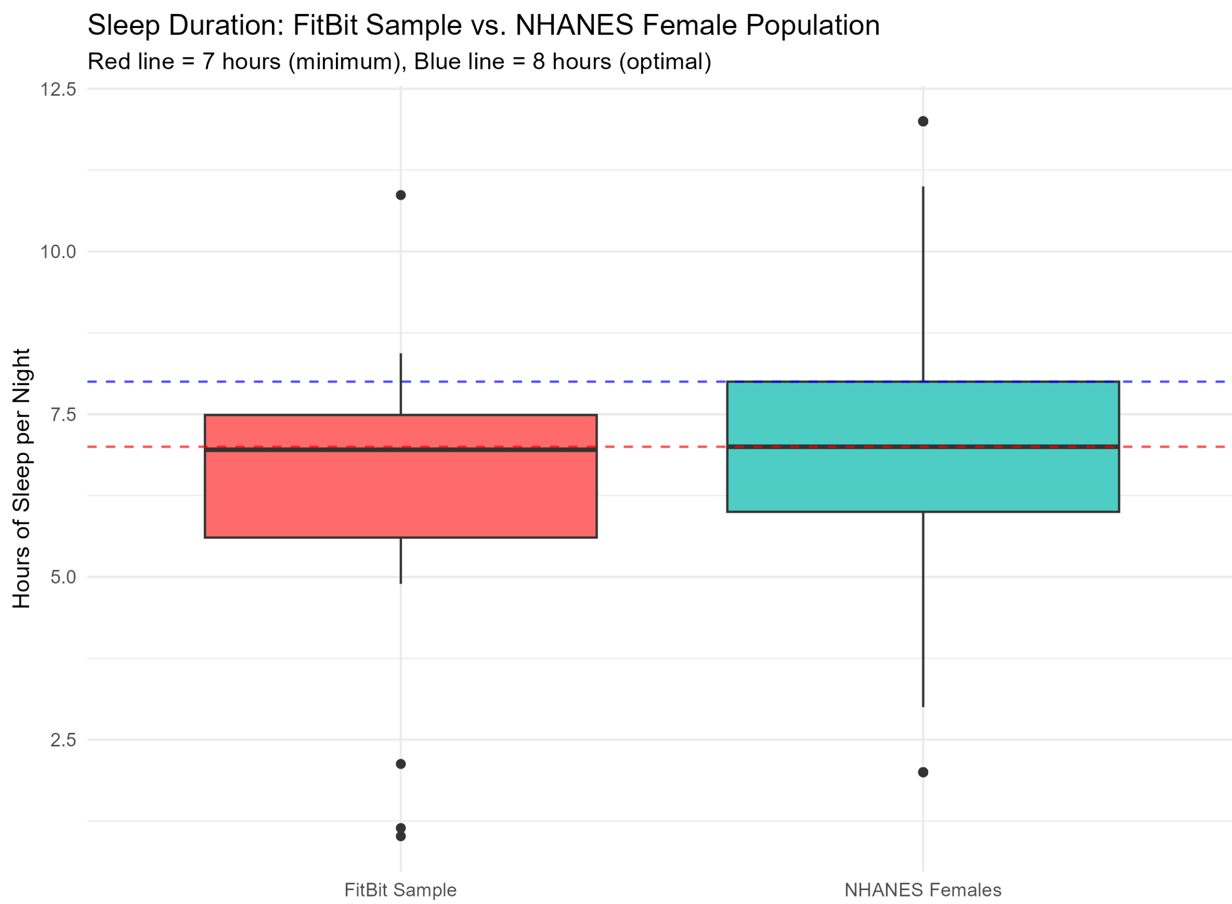


Figure 3: Comparison plot - Sleep Duration: FitBit Sample vs. NHANES Female Population

Comparison Results:

- FitBit Sample: Median = 6.95 hours, Mean = 6.29 hours
- NHANES Females: Median = 7.00 hours, Mean = 6.99 hours
- Statistical test: p-value = 0.146 (no significant difference)

Interpretation: Our FitBit sample's sleep patterns closely align with the general US female population, suggesting our findings are generalizable despite the small sample size.

BMI Comparison

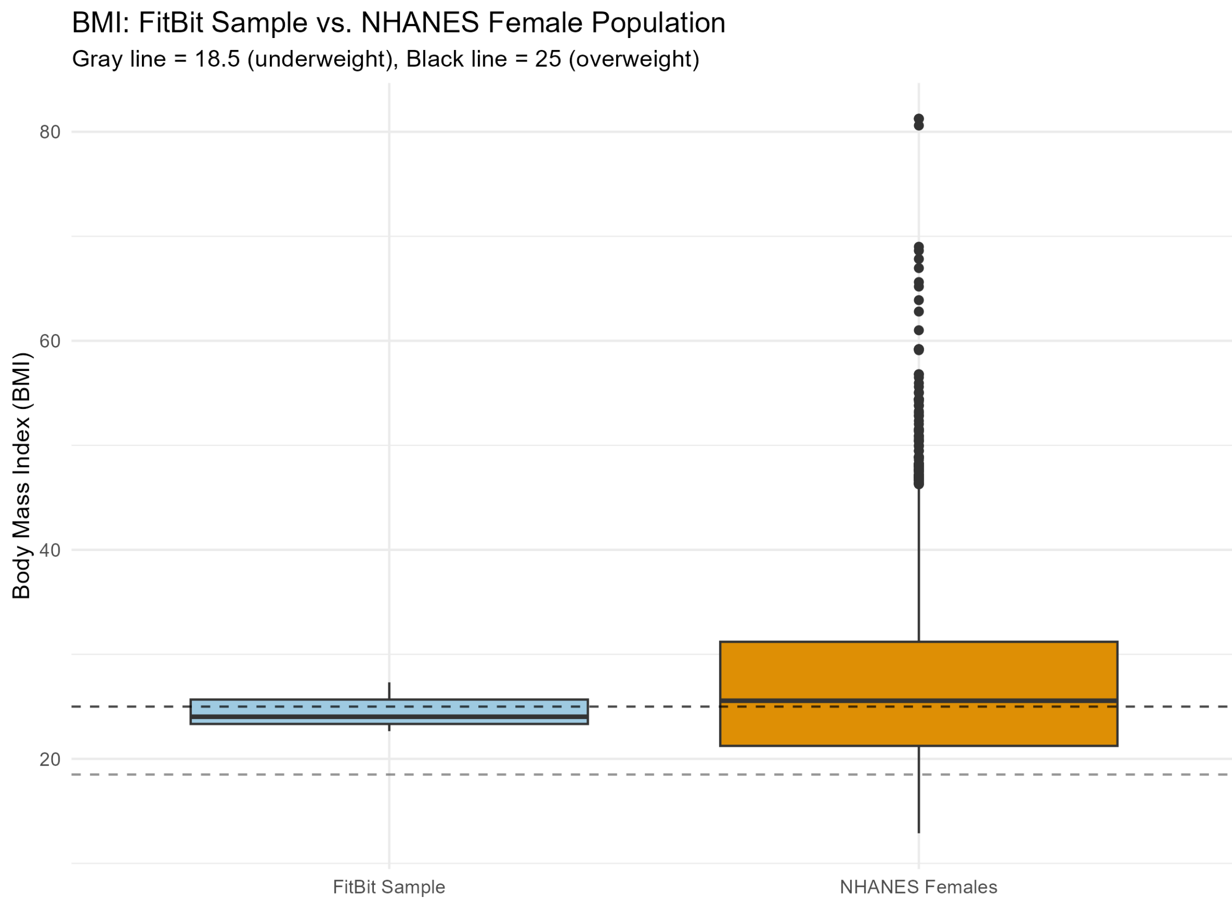


Figure 4: BMI Comparison Plot - colorblind-friendly version

Comparison Results:

- FitBit Sample: Median = 24.03 (n=3 with BMI data)
- NHANES Females: Median = 25.56 (n=4,841)

Note: Limited BMI data in FitBit sample (only 3 participants) prevents strong conclusions, but available data suggests similar body composition to national average.

Click here to view Comparison summary table

Validation Conclusion

The close alignment in sleep duration and BMI between our FitBit sample and NHANES data suggests that despite the small sample size, **our participants are representative of typical US women**, not extreme fitness enthusiasts. This strengthens the case that the sleep-activity trade-off we discovered is a real phenomenon affecting everyday women.

Bellabeat Marketing Strategy Recommendations

Based on the analysis revealing that active women sacrifice sleep for activity, here are strategic recommendations for Bellabeat:

Core Positioning

From: Traditional fitness tracker celebrating "more is better"
To: Holistic wellness coach promoting "balance over burnout"

Key Message: *"Being Active Shouldn't Cost You Your Sleep - Bellabeat Helps You Balance Both"*

Target Audience

Primary: "The Busy Overachiever"

- Demographics: Women 25-45, working professionals, health-conscious
- Behavior: High daily steps (8,000-12,000), poor sleep (<6-7 hours)

- Pain point: "I'm exhausted but can't stop moving"
- Value proposition: "Sustainable wellness without burnout"

Product Feature Recommendations

1. Balance Score

- Combine activity AND sleep into one wellness metric
- Alert users when one metric is suffering: "Warning: High activity but low sleep = burnout risk"
- Recommendation engine: "You've been active 5 days with <6 hours sleep. Consider a rest day."

2. Sleep Efficiency Tracking

- Show time awake in bed (current finding: ~14% of time in bed is spent awake)
- Personalized insights: "You spent 45 minutes awake in bed - try these sleep tips"
- Evening wind-down alerts when late activity may impact sleep

3. Recovery Mode

- When sleep debt accumulates, suggest gentler activity
- Permission to rest: "Your body needs recovery. Try yoga instead of intense exercise today."

Campaign Concept: "The Sleep-Activity Paradox"

Headline: "You're crushing your step goal. So why are you exhausted?"

Body Copy: "New data reveals what many active women already know: the busier and more active you are, the less you sleep. But your body needs both movement AND rest to thrive. Bellabeat tracks the balance, not just the numbers. Because wellness isn't about choosing between being active or being rested—it's about achieving both."

Differentiation from Competitors

Traditional Fitness Trackers	Bellabeat
Celebrate only activity	Celebrate balance
“More steps = better”	“Right balance = better”
Guilt for rest days	Permission to rest
Activity-focused	Holistic wellness

Supporting Data Points for Marketing

- “Women sleeping less than 6 hours take 50% more steps than those sleeping 8+ hours - but at what cost?”
- “The most active days correlate with the worst sleep - your body is trying to tell you something”
- “99.3% of our users successfully track both sleep and activity when engaged - proving balance is possible”

Study Limitations & The COVID-19 Context

Temporal Limitations

Most Significant: This data was collected in **April-May 2016**, representing pre-pandemic behavioral patterns. The COVID-19 pandemic (2020-present) fundamentally altered:

- **Work patterns**: Remote work eliminated commutes but blurred work-life boundaries
- **Exercise habits**: Gym closures drove outdoor activity surge; some people exercised more, others less
- **Sleep patterns**: Stress, screen time, and schedule flexibility all changed
- **Mental health**: Anxiety, isolation, and uncertainty affected wellness behaviors

Implication: The sleep-activity trade-off identified here may no longer accurately represent current user behavior. The findings serve as a valuable **pre-COVID baseline** but require validation with post-2020 data.

Sample Size

- Only 24 participants with matched sleep/activity data
- Small sample increases uncertainty, though NHANES validation suggests representativeness *for 2016 population*

BMI Data

- Only 3 participants logged weight consistently
- Limits our ability to analyze relationships between body composition, activity, and sleep

Self-Selection Bias

- FitBit users may be more health-conscious than general population
- Though NHANES comparison suggests similarities, findings may not apply to completely inactive individuals

Future Research Directions

Priority #1: Pre/Post-COVID Comparative Analysis

Research Question: How did the COVID-19 pandemic alter the sleep-activity trade-off identified in this 2016 data?

Methodology:

1. Obtain equivalent FitBit or fitness tracker data from 2020-2024
2. Match demographic profiles (age, gender, activity level) to 2016 sample
3. Compare correlation coefficients between time periods
4. Segment by remote work status, parental status, and occupation type
5. Identify which groups improved balance vs. worsened trade-off

Expected Value:

- Understand how external disruptions affect health behaviors
- Identify vulnerable user segments
- Inform crisis-resilient product design
- Guide messaging for post-pandemic wellness priorities

Additional Research Directions

1. **Longitudinal study**: Track individual users over 6-12 months to see if sleep-activity trade-off persists or self-corrects
2. **Intervention testing**: Do "balance alerts" actually help users improve both sleep and activity?
3. **Demographic segmentation**: How does the trade-off vary by age, occupation, parental status, or life stage?
4. **Qualitative research**: Interview users about *why* they sacrifice sleep—time constraints? Cultural pressure? Lack of awareness?

Why This Matters Professionally

The COVID-19 pandemic created a natural experiment in human behavior change. Companies like Garmin, Nike Run Club, and Runkeeper have multi-year datasets that span this transition—and the insights from analyzing that data could reshape how we think about wellness technology for a post-pandemic world.

This is exactly the kind of research I'm passionate about pursuing as a data analyst: using rigorous methods to understand how people's lives have changed, then translating those insights into products and strategies that genuinely help.

Conclusion

This analysis reveals a critical insight for Bellabeat's marketing strategy: **active women are trading sleep for steps**, with the most physically active users being the most sleep-deprived. This unsustainable pattern presents a unique market positioning opportunity.

Key Findings:

1. Negative correlation between sleep and activity ($r = -0.19$, $p < 0.001$)
2. Users sleeping <6 hours take ~50% more steps than those sleeping 8+ hours

3. FitBit sample is representative of general US women (validated via NHANES)
4. Sleep efficiency gap exists - users spend ~14% of time in bed awake

Strategic Opportunity:

While competitors celebrate maximum activity, Bellabeat can own the "balance and sustainability" position. By helping women achieve both adequate sleep AND healthy activity levels, Bellabeat differentiates itself as the anti-burnout wellness partner.

Personal Reflection: As someone who relates deeply to these findings—juggling half marathon training with work and family—I recognize the need for technology that encourages balance, not just achievement. Bellabeat has the opportunity to be that voice of sustainable wellness that active women desperately need.

The Unanswered Question: What Happened During COVID?

This analysis captures a snapshot of pre-pandemic life in 2016. But the world has changed dramatically:

Hypotheses Worth Testing with Post-COVID Data:

Scenario 1: Remote Work Improved Balance

- Eliminated commutes gave people time for both sleep and exercise
- More flexible schedules allowed for midday workouts without sacrificing evening sleep
- **Prediction:** Negative correlation weakens; people achieve both sleep and activity

Scenario 2: Remote Work Worsened the Trade-Off

- Work-from-home blurred boundaries, extended working hours into former sleep time
- Increased screen time and "Zoom fatigue" worsened sleep quality
- Stress-driven exercise (running as escape) maintained high activity despite worse sleep
- **Prediction:** Negative correlation intensifies; trade-off is more severe

Scenario 3: Divergent Paths

- Some segments thrived (professionals with flexible schedules, no kids)
- Others struggled (parents homeschooling, essential workers with longer hours)
- **Prediction:** Bimodal distribution emerges; one group finds balance, another burns out harder

The burning question: Has the sleep-activity trade-off I identify in this pre-COVID data intensified, reversed, or evolved entirely in our post-pandemic world? Did remote work give people more time for both sleep and exercise, or did the blurred boundaries make the trade-off worse?

I would welcome the opportunity to bring this analytical approach to organizations like **Garmin**, **FitBit**, **Apple**, **Whoop**, **Amazfit**, **Nike** or new start-up as a data analyst, specifically to conduct comparative analyses between pre- and post-COVID user behavior. By examining more recent data alongside this 2016 baseline, we could identify:

- How activity patterns shifted when gyms closed and running became a primary outlet
- Whether remote work improved sleep-activity balance or worsened it
- Which user segments thrived vs. struggled during the pandemic
- What lasting behavioral changes should inform product development and marketing

The Research Opportunity:

Comparing this 2016 baseline against 2020-2024 data would reveal:

- How global events reshape health behaviors at scale
- Which user segments are most vulnerable to burnout
- What features help users maintain balance during disruption
- How to design resilient wellness technology for an uncertain world

This is precisely the type of longitudinal, behavior-change analysis that excites me as a data analyst. Organizations like **Garmin**, **Nike Run Club**, **Apple**, **FitBit** and **Runkeeper** sit on treasure troves of multi-year user data that could answer these questions—and I would be thrilled to help uncover those insights.

Final Thoughts

This case study demonstrates that even with limited data (n=24), rigorous analysis can uncover meaningful patterns and generate actionable business insights. Imagine what we could learn with:

- Larger sample sizes
- Longer time horizons
- Multi-year comparative data (pre/during/post-COVID)
- Demographic segmentation (age, occupation, parental status)
- Intervention testing (do "balance alerts" actually work?)

The questions are endless. The potential impact is significant. The opportunity to contribute to this field—whether at Bellabeat, Garmin, Nike, Runkeeper, Apple or elsewhere—is what drives my passion for data analytics in the wellness space.

If you're working on similar analyses or interested in discussing this research, I'd love to connect. Because at the end of the day, we're not just analyzing datasets—we're trying to help real people live healthier, more balanced lives.

And in 2026, after everything we've been through collectively, that mission matters more than ever.

Analysis completed: [01/29/2026]
Data source: FitBit Fitness Tracker Data (April-May 2016) via Kaggle
Validation: NHANES 2009-2011 National Health Data
Tools: R (tidyverse, ggplot2), R Markdown